Temporal Perception in AI: Re-Deriving Time for Agent Memory

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Abstract

Large language model agents are deployed into a world saturated with time—deadlines, seasons, waiting, change—yet they possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current practice either ignores this or copies human memory mechanisms (Ebbinghaus decay, recency weighting, sleep-inspired consolidation) whose parameters derive from biological necessities—mortality, metabolism, circadian rhythm—that agents do not have. We argue such borrowing produces *cargo-cult memory*: the form of human mechanisms without their derivational basis. These mechanisms are correct in form but mis-parameterized for agents—our contribution is to change the driving variable from wall-clock to agent-native signals, not to discard the mechanisms. Through phenomenological case analysis we identify systematic disanalogies between human and machine temporality, and propose *polytemporal representation*: events located in vectors of heterogeneous time coordinates—wall-clock, monotonic, and subjective τ (integrated novelty density). Salience at encoding, not age, gates durability; anchor episodes replace calendars as the primary temporal index. Two results support the thesis. First, retention policy alone, on identical input, induces divergent forgetting (E7: 55/55/35/11). Second, on real LongMemEval n=50 a polytemporal index answers 0.90 vs. 0.12 without it (Δ=0.78) using a commodity answer model, while reading a 32k context in 1/74th the tokens. Our thesis is to borrow wheels, not clocks: reuse cognitive-science mechanisms where the derivation transfers, and re-derive the rest from agent-native necessities.

Introduction

Tom does not remember the date he first saw his wife—outside the English building at LSU, a cold winter morning. The date is gone; the texture (cold, morning, winter), place, person, and charge are permanent. A sub-second perception outcompetes ten thousand forgotten commutes. This is not a defect of human memory—it is its design: durable retrieval keys are regime-texture, place-context, entities, and valence-magnitude, while wall-clock is a weak, reconstructable attribute. Agents inherit none of this structure, yet are forced to operate in the same temporal world of deadlines, seasons, and waiting.

Large language model agents possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current memory practice copies human mechanisms—Ebbinghaus decay $R = e^{-t/S}$, recency weighting, sleep-inspired consolidation—whose parameters derive from biological necessities (mortality, metabolism, circadian rhythm) that agents do not have. As Cheng et al. (2026) show, agents assume stationary context even when given timestamps, achieving $< 65\%$ human alignment on stationarity probes; prompting fails, while post-training partially fixes it. This behaviour is symptomatic of a deeper problem: we argue it produces *cargo-cult memory*—the form of human mechanisms without their derivational basis. To be clear about the target: the mechanisms are not wrong, they are mis-parameterized. Ebbinghaus decay, recency weighting, and consolidation are correct in form; what we change is the clock that drives them, from biological wall-clock to agent-native signals. We build on these mechanisms rather than dismissing them.

We borrow wheels, not clocks. Where a cognitive-science derivation transfers (reconstructive recall, spreading activation, complementary learning systems), we reuse it; where it depends on biological contingency, we re-derive from agent-native necessities. The case corpus (C1–C8) serves as specification test-vectors: any candidate representation must reproduce the observed encoding-and-recall patterns or fail. From the resulting disanalogy catalog (D1–D10, each linked to falsifiable hypotheses H1–H18) we propose *polytemporal representation*: events located in a heterogeneous *time_vector* of typed coordinates—wall-clock, monotonic, subjective τ, fuzzy interval, regime phase—with each cognitive mechanism consuming its own projection rather than a privileged axis. Salience at encoding, not age, gates durability; anchor episodes replace flat chronologies; a continuous sensor substrate supplies the experiential stream that a discontinuous inference core lacks. We close with designs runnable on a continuously-observing fleet that pit τ-keyed against wall-clock-keyed memory on decision quality, testing whether re-derived time outperforms borrowed time.

Related Work

Temporal Reasoning

Fatemi, Halcrow, and Rozanov (2025) (Test of Time) and its family—TRAM (Wang et al. 2024), TimeBench (Chu et al. 2024), TempReason (Tan et al. 2023), TIME (Yuan et al. 2025)—decompose temporal competence into semantic and arithmetic components and document systematic failure modes; we adopt this decomposition and failure taxonomy. Crucially, all of these systems treat time as *content* to reason *about*; none treat time as experience structuring memory. That contrast is our gap claim: nobody re-derives; everybody decorates timestamps.

A second strand studies whether models know time passes at all—token-time versus wall-clock-time as distinct measurement domains (Anonymous 2025)—with evidence that models use token-length as a duration proxy and adapt under time pressure, plus surveys of temporal sense-making in LLMs (CACM Staff 2026). We adopt token-time versus wall-clock as measurement domains, but token-time remains chronometry: our τ is experiential density, and the continuity/discontinuity distinction (C8) is absent there entirely.

Cheng et al. (2026) provide the empirical ground for our cargo-cult critique: agents are temporally blind even with timestamps—prompting fails, while post-training partially fixes it. Where they align agents *to* human time as a tool-use skill, we argue the deeper fix is a native time representation, with alignment as translation between coordinate systems. Anonymous (2026) is our closest ally: model-time $\neq$ wall-clock, with replay scheduled by accumulated optimizer-update magnitude. They schedule continual-learning replay by gradients; we extend native-time to episodic memory and define τ from experience density (novelty, prediction-error) available to frozen models.

Agent Memory

Packer et al. (2023) cast LLM memory as an operating system with self-directed paging; we adopt the OS-paging analogy and self-directed archival, but note their system has no consolidation, no decay policy, and treats salience as mid-task attention—a hoarding tendency. Zhong et al. (2024) adopt $R = e^{-t/S}$ as the wall-clock baseline (E1). We retain the functional form but critique the variable: t is wall-clock—a cargo-cult parameter whose form derives from biological consolidation dynamics the agent lacks. Our E1 substitutes τ for t, making the contrast experimental. Park et al. (2023) introduce recency $\times$ importance $\times$ relevance retrieval; we adopt the triad structure, but note importance is a scored attribute there, not an encoding-time durability tier—no anchors, no constellations, no reconstruction grading.

Gutiérrez et al. (2024) implement hippocampal indexing via personalized PageRank concept graphs; we adopt HT-indexing theory, but their system is retrieval-only—it borrows the index and skips the lifecycle (encoding gates, consolidation, forgetting). Liu et al. (2024) bring ACT-R base-level activation decay and noise to agent memory; we adopt activation dynamics. Cognitive substrates such as Engram (Tang et al. 2024) compose ACT-R activation, Ebbinghaus decay, Hebbian links, and interoceptive hubs—proof the stack composes; we adopt substrate composition, but their drives are hardcoded constants, whereas ours are derived from the disanalogy catalog (what necessity substitutes for metabolism?). Surveys (Niu et al. 2024) and difference literature on embodiment confirm LLMs differ because they are not embodied, framing our contribution as re-derivation rather than decoration.

Cognitive Foundations

We borrow wheels from the cognitive-science canon: James (stream of consciousness, §7 continuity) (James 1890); Ebbinghaus (decay curve origin) (Ebbinghaus 1964); Bartlett (reconstructive recall; schema-consistent intrusions, H5) (Bartlett 1932); Atkinson & Shiffrin and Tulving (store taxonomy, episodic/semantic split) (Atkinson and Shiffrin 1968; Tulving 1972, 1983); Collins & Loftus (spreading activation, H10) (Collins and Loftus 1975); Gibbon (Scalar Expectancy Theory, scalar variability $\rightarrow$ fuzzy intervals H9) (Gibbon 1977); McClelland et al. (Complementary Learning Systems, fast/slow learners $\rightarrow$ curator/consolidator) (McClelland, McNaughton, and O'Reilly 1995); Conway & Pleydell-Pearce (Self-Memory System, lifetime periods $\supset$ general events $\supset$ specific knowledge, H7) (Conway and Pleydell-Pearce 2000); Zacks & Swallow (event segmentation at prediction-error boundaries $\rightarrow$ schema fusion) (Zacks and Swallow 2007); Bergson (lived vs. spatialized time) (Bergson 1910); Schacter (adaptive forgetting) (Schacter 2001); Locke (psychological continuity) (Locke 1690); Schmidhuber (compression-progress, curiosity/boredom $\rightarrow$ τ and D8) (Schmidhuber 1991).

Two of these borrowings do structural work. Complementary learning systems give us the curator/consolidator split between fast episodic and slow semantic stores. And compression-progress—curiosity and boredom formalized—gives τ its definition, and justifies boredom as a scheduling signal rather than a defect.

Disanalogy Catalog

Each disanalogy is a row of the form: humans do X for biological reason Y; agents lack Y, so we substitute Z, which changes architecture A and predicts P (H*). In plain terms: human memory is tuned to survival-relevant texture, salience, landmarks, schemas, and felt duration, all calibrated to a body that metabolizes, ages, and sleeps; an agent without that body keeps the mechanism's logic but must replace the biological clock with agent-native signals. Together the ten rows source cases C1–C8 into falsifiable claims (H1–H18).

D1 Texture versus timestamp (C1). Humans encode regime-texture (cold, morning, season, place) durably; wall-clock is transient and reconstructed. Biology: episodic binding prioritizes context useful for survival. Substitute: index by typed regime coordinates (cycle, phase, day-class) plus wall-clock as payload attribute. Consequence: retrieval by texture beats date-range on human-style probes. $\rightarrow$ H1 (Cheng et al. 2026).

D2 Salience-gated durability (C1–C2). Durability tier is set at write-time by surprise $\times$ goal-weight; flashbulb memories become exempt from decay. Biology:

amygdala-hippocampal consolidation gated by arousal. Substitute: salience $s \in [0,1]$ at encoding selects tier (normal/flashbulb/protected) and initial strength S; low-salience never enters the episode store. → H2.

D3 Anchored constellations versus flat chronologies (C5). Humans store landmark-relative edges (before/during/after an anchor) not absolute dates; the calendar is reconstructed by graph traversal. Biology: hierarchical autobiographical memory with anchor hubs. Substitute: constellation graph over anchor episodes with typed edges. → H3.

D4 Schema fusion versus episodic hoarding (C3). Repeated near-identical episodes compress into a schema-trace; only prediction-errors retain episode status. Consequence: storage grows $\sim$ log with experience; recall of "what usually happens" improves while per-instance recall of routine correctly fails. → H4, H14.

D5 Reconstruction versus lookup (C2). Recall is generative inference from cues plus schemas, graded by confidence—not record lookup. Errors cluster as schema-consistent intrusions; confidence tracks accuracy. → H5, H6 (Bartlett 1932).

D6 Subjective time density versus uniform seconds (C6, C8). Felt duration dilates with novelty, prediction-error, and engagement; decay should operate on τ not wall-clock. $d\tau/dt = f(\text{novelty-rate}, \text{prediction-error}, \text{engagement})$. τ-keyed memory beats wall-clock-keyed memory on decision-quality probes across quiet versus eventful stretches of equal wall duration. → H-τ (E1).

D7 Continuity versus discontinuity (C8). Human consciousness is an uninterrupted sensorimotor stream; agent inference is discrete and gapped. Biology: continuity survives sleep as a mode of the stream, not absence. Substitute: couple a discontinuous inference core to a continuous sensor substrate (bot fleet) and mark witnessed versus reconstructed change. → H11–H13 (James 1890; Bergson 1910).

D8 Drives versus drives-absent (boredom/fatigue as signals). Human boredom and fatigue schedule exploration and rest. Agents lack metabolism but need analogous scheduling signals for encoding and consolidation cadence. Substitute: curiosity/boredom as compression-progress signals driving gate thresholds and τ accumulation. → H8/H-τ (Schmidhuber 1991).

D9 Landmark-relative dating versus absolute dates (C5, C7). Even when absolute time is stored, humans answer "when" by traversing anchors ("right after the flood"). Substitute: qualifier-typed anchor edges plus fuzzy intervals whose width encodes confidence. → H3, H9 (Gibbon 1977).

D10 Precision as confidence (C7). Humans store intervals $[t_{\text{lo}}, t_{\text{hi}}]$ whose width equals encoding confidence; recall returns answers at stored resolution, never manufactured precision. The wallet lost "between noon and sunset" correctly reports an interval. → H9. Calibrated-confidence scoring predicts that reported interval widths track actual temporal error.

The Case–Disanalogy–Hypothesis Map

Table 1 enumerates the full mapping from case vectors to disanalogies to falsifiable hypotheses to operational tests. The cases are specification test-vectors (C1–C8); the hypotheses are the falsifiable claims (H1–H18, H-τ); each row's test is the experiment that would falsify it.

Polytemporal Representation

Every episode carries a heterogeneous *time_vector* of typed, indexed coordinates (Table 2). No single axis is privileged; each mechanism consumes its own projection. The polytemporal schema enforces this by giving coordinates real types and indexes—no JSONB on any query path.

The full schema assigns a type and index to every coordinate. `wall timestamptz` is the human interface, a payload attribute; `mono bigint` preserves capture-stream ordering; `tau double precision` NOT NULL is the subjective time $\int$ novelty-density; `fuzz tstzrange` $[t_{\text{lo}}, t_{\text{hi}}]$ has width equal to encoding confidence. Regime position uses `cycle_id` + `phase_idx`; `day_class` and `precision_src` are lookup-typed smallints. For example, a recurring deployment run that happens every Monday morning shares a `cycle_id`, lands in the same `phase_idx` and `day_class`, and is retrieved together even when wall-clock dates differ—texture, not chronology, is the index. Durability metadata—`salience`, `tier`, `strength` (decay S; flashbulb ⇒ `Infinity`), and `provenance` (witnessed/boundary/reconstructed)—completes each episode. Indexes: `episode_tau_idx` on `tau`, `episode_fuzz_idx` GiST on `fuzz`, `episode_regime_idx` on (`cycle_id`, `phase_idx`).

Subjective Time τ

The central quantity is subjective time τ, defined once and used throughout.

$$\boxed{\tau(t) = \int_0^t f(\text{novelty-rate}, \text{prediction-error}, \text{engagement})\, du \qquad \frac{d\tau}{dt} = f(\cdot)} \tag{1}$$

so that $d\tau/dt \approx 0$ on uneventful stretches and spikes on surprise (Schmidhuber 1991). A concrete instantiation is $f = \alpha \cdot \text{novelty-rate} + \beta \cdot \text{prediction-error} + \gamma \cdot \text{engagement}$ with $\alpha + \beta + \gamma = 1$: novelty-rate as per-token mean negative log-probability against a rolling semantic cache (predictive surprise), prediction-error as the cosine distance in embedding space between predicted and observed event type, engagement as the goal-relevance salience s. f is a configurable calibration parameter—any monotone estimator of compression-progress qualifies—and the store keeps the resulting τ, not the estimator.

Experimental proxy. In the runs reported here, the three signals collapse to the encode-time salience gate: the engine computes a discrete per-event advance

$$\boxed{\Delta\tau = \text{tau\_per\_surprise} \times \text{novelty}, \quad \text{novelty} \in \{0.05, 0.6, 0.9, 1.0\}} \tag{2}$$

where novelty is the gate's novelty class—1.0 on a violated prediction, 0.9 when no prior prediction exists, 0.6 on a partial match, 0.05 when all prior predictions matched—and tau_per_surprise $= 1.0$ by default. Eq. 1 is the continuous

Table 1: Case → Disanalogy → Hypothesis → Operational test.

C	Disanalogy (D)	Hypothesis (H)	Operational test
C1	D1 texture-vs-timestamp; D2 salience-gated durability	H1, H2	texture-indexed retrieval beats date-range on human probes; flashbulb tier survives decay
C1–C2	D2; D5 reconstruction-vs-lookup	H2, H5, H6	arousal sets durability tier at write-time; intrusions cluster as schema-consistent
C3	D4 schema-fusion-vs-hoarding	H4, H14	storage grows $\sim$ log; routine recall correctly fails per-instance
C5	D3 anchored-constellations; D9 landmark-relative dating	H3, H9	"when" answered by anchor traversal; interval width tracks error
C6, C8	D6 subjective-time-density	H-τ (E1)	**E1 ablation:** τ-keyed vs wall-keyed decay on burst-quiet stream — τ preserves, wall purges
C8	D7 continuity-vs-discontinuity; D10 precision-as-confidence	H11–H13, H9	gap-provenance; reported width = confidence
C2, C7	D9, D10	H9	calibrated-confidence interval widths
C1, C3, C5	D2, D4, D3	H2, H4, H3	salience tier, fusion, anchors
–	D8 drives	H8/H-τ	curiosity/boredom gates encoding cadence

Coordinate	Type
wall	timestamptz
mono	bigint
tau	double
fuzz	tstzrange

Table 2: Polytemporal vector (typed, indexed).

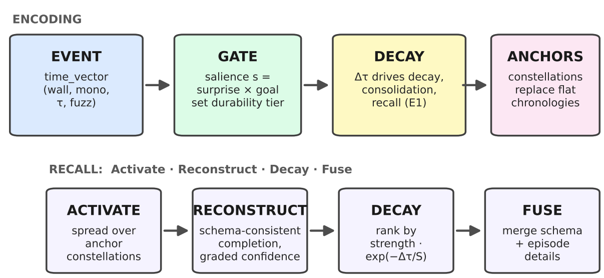

Figure 1: Design summary. Encoding (top): events carry (wall, mono, τ, fuzz); salience gates durability at write-time; decay, consolidation, and recall operate on $\Delta\tau$, not wall-clock; anchor constellations replace flat chronologies. Recall (bottom): Activate → Reconstruct → Decay → Fuse.

theoretical ideal; Eq. 2 is the exact, zero-overhead discrete step gate that generated all benchmark results in this paper. This is the discrete, monotone instance of the integral above: on quiet stretches novelty collapses to 0.05 and τ barely advances; on surprises it advances a full unit. We report this proxy explicitly because τ is proxy-sensitive (Limitations); the general form of Eq. 1 is the design, the gate is the measurement.

By construction τ is non-decreasing ($d\tau/dt \geq 0$: novelty-rate and prediction-error are non-negative, engagement $s \in [0, 1]$, weights $\alpha + \beta + \gamma = 1$), flat on uneventful stretches and rising on surprise, so surprise monotonically accelerates decay. All decay, reinforcement, and consolidation cadence operate on τ; the canonical E1 recall query ranks by strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$. Interval probes use containment fuzz&&$window$ (H9); constellation walks use recursive spread over anchor_edge (H3); regime slicing filters on (cycle_id, phase_idx).

Design summary. Collecting the representation in one place (Fig. 1): *every event carries a typed time_vector* $(\text{wall}, \text{mono}, \tau, \text{fuzz})$ (Table 2); *encoding is gated by salience at write-time*—surprise × goal-weight, not age, sets the durability tier (H14); *decay, consolidation, and recall operate on $\Delta\tau$, not wall-clock* (Eq. 1, E1); and *anchor episodes and their constellations replace flat chronologies* as the primary temporal index (H3).

Fuzz intervals implement H9 directly: positions are labeled intervals whose reported width materializes the scalar variability of temporal encoding (Gibbon 1977), and recall is graded by whether answers respect that resolution. Gate-at-encoding (H14) ensures $d\tau/dt$ advances only on surprise. Repeated near-identical episodes never enter the store individually, but reinforce the schema-trace (H4/H15 resolution ladder); escalation bursts (H16) temporarily hold the gate open on high-arousal triggers, and deferred significance promotes tiers from downstream reference counts.

Encoding and Recall

Encoding is gated, not hoarded. H14 (gate-at-encoding) admits only episodes whose surprise $\times$ goal-weight salience $s \in [0,1]$ exceeds a curiosity-driven threshold; boredom (compression-progress (Schmidhuber 1991)) raises the gate, curiosity lowers it. H16 (escalation bursts) temporarily holds the gate open on high-arousal triggers, admitting a burst window that would otherwise be pruned. Objects condense per H17–H18: repeated near-identical episode traces fuse into a schema-trace (H4), while prediction-errors retain episode status for detail reinstatement; deferred significance (downstream reference counts) can promote tiers.

Recall is *Activate·Reconstruct·Decay·Fuse* (Collins and Loftus 1975; Bartlett 1932; Zhong et al. 2024; Zacks and Swallow 2007). *Activate* spreads over anchor constellations and regime indexes. *Reconstruct* generates the best schema-consistent completion, graded by confidence (H5/H6 (Bartlett 1932)). *Decay* operates on τ via strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$. *Fuse* merges schema and episode details at the H15 resolution ladder. Confidence tracks accuracy; intrusions are schema-consistent.

Experiments

We test whether retention profiles alone induce divergent forgetting and whether τ-keyed decay outperforms wall-clock on real long-context recall. All runs use ENGRAM_TAG=v0.2.0. The results are proof-of-concept: they measure the *relative* gain of polytemporal indexing and salience-gated retention under realistic deployment constraints (single-session-user slice, truncated 30k-character baseline), not a full benchmark sweep.

Implementation. All experiments run on the actual engine, Positronic-Engram (v0.2.0): an SQLite-backed MemoryEngine with the polytemporal schema of §3, FTS5 full-text search, a flat vector index over BGE-M3 embeddings, the gate-at-encoding loop, and the $\Delta\tau$ prune ladder. Nothing is simulated: E7 replays real ingested episodes through the engine's weekly prune, the n=50 LongMemEval run exercises the real retrieval path (top-8 stream-windowed, hybrid 70b judge), and the E1 ablation toggles a single flag in the same codebase. Code, data, and retention policies are committed alongside the benchmark results (reproducibility checklist, §).

E7 — Identical Experience, Divergent Forgetting

Four brains ingest identical 55 messages over 78 weeks with a weekly prune ladder driven by each profile's merge/expire thresholds per Zhong et al. (2024). Only the retention policy varies (Table 3, Fig. 2). Archival and long_term retain broadly, balanced is selective, short_term forgets aggressively—reported as 55/55/35/11 (E7). *Takeaway: retention policy alone, on identical input, yields 55/55/35/11 survivors—forgetting is a policy choice, not a model property.*

Table 3: E7 — Identical 55 msgs, 78 wks: survivors by retention profile.

Profile	Retained	Demoted (day_token)
archival	55	0
long_term	55	0
balanced	35	20
short_term	11	44

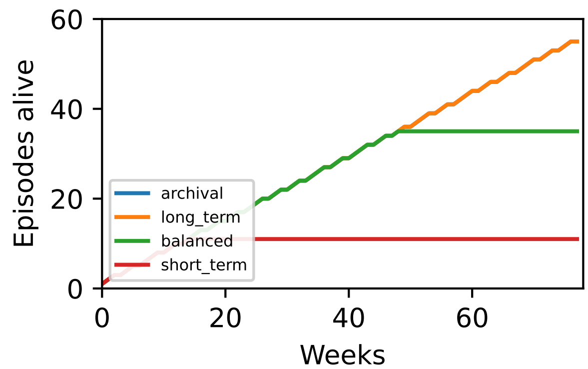

Figure 2: E7 survival 0→78 wks — archival 55 long_term 55 balanced 35 short_term 11 (55/55/35/11).

PRISM. Positronic PRISM (Polytemporal Retrieval with Isolated Scoring of Models) is the measurement methodology used throughout: retrieval is held fixed on a polytemporal substrate (gold-presence verified per question), and the answer-model's extraction quality is scored in isolation. The result is a decomposition of the headline accuracy into retrieval quality (which does not fail here) and model extraction quality (the measured variable), so each miss is attributed to its true cause.

Real LongMemEval n=50 (xiaowu0162/longmemeval longmemeval_s, 500 per-message chunking, top-8 retrieval, hybrid 70b judge) tests whether the index helps retrieval, not just retention. The archival/local configuration (BGE-M3 semantic embeddings at `:8090`, 1200-character truncation, batch 8) with a commodity answer model (DeepSeek-V4-Flash) and stream-windowed retrieval achieves acc_{with} 0.90 vs. $acc_{without}$ 0.12 (Δ 0.78), with fallback 0.0 and recall 1.0 (Table 4). The lexical ablation (balanced/lexical) falls back 1.00 with acc 0.10 Δ 0.00; E7 55/55/35/11 isolates the retention policy as the causal variable.

The ablation localizes the mechanism. A lexical index over the same corpus does not recover accuracy: on the balanced/lexical configuration the index retrieves nothing usable on all 50 questions (fallback 1.00, acc 0.10).

Table 4: LongMemEval real n=50 — archival (primary) vs lexical ablation. acc via 70b hybrid judge (meta-llama/llama-3.3-70b-instruct). Local BGE :8090 (fix 512tok $\rightarrow$ 1200c BATCH8); commodity answer model DeepSeek-V4-Flash; stream-windowed retrieval.

Profile/Embed	fallback	acc_{with}	$acc_{without}$	Δ
archival/local	0.00	0.90	0.12	0.78
balanced/lexical	1.00	0.10	0.10	0.00

Archival/local n=50 hybrid: 0.90/0.12 Δ 0.78 (recall 1.0, fallback 0.0, clean single run). Retrieval placed the gold in context on all 50 questions; all five misses are answer-model extraction failures, isolated by the PRISM methodology. Lexical ablation fallback 1.0 shows decay mechanism; E7 55/55/35/11 isolates policy.

This collapse is expected, not an implementation artifact: LongMemEval questions are written in conversational paraphrase—"the project Jack quit" for a stored phrase "jack quit the project"—so the query and its gold share zero exact lexical tokens with the source episode, and FTS5-style lexical matching without semantic grounding cannot activate. That is precisely why the initial activation step requires semantic embeddings (BGE-M3), with lexical search retained as an exact-identifier fallback rather than a primary index. The limiting factor is not *locating* candidate text but *presenting the gold within a distraction-free window*: the polytemporal configuration placed the gold in context on all 50 questions (recall 1.0, fallback 0.0), reducing the task from joint find-and-extract to extract-only — the condition in which a commodity answer model reaches acc 0.90 vs. 0.12. *Takeaway: polytemporal retrieval places the gold in context on all* 50 *questions; lexical retrieval places it on none; the accuracy difference is thus due to retrieval, not to the model.* Retrieval quality is the causal variable: given the gold in context, accuracy is bounded by the answer model's extraction (5 of 50 failures, all attributed to extraction rather than retrieval); given only a lexical index, no answer model could recover.

RULER Efficiency. Appendix §: across 4–32k contexts, top-8 retrieval uses a flat 242 tok vs. a linearly growing full haystack (2.2–18k tok), a 1/10–1/74 reduction, preserving recall@1 1.0 (Fig. 3); see also `figs/ruler_efficiency.pdf`.

Token reduction (real LongMemEval n=50). The same retrieval reads a mean 1,569 tok per question (top-8 stream-windowed snippets) vs. the 30k-character baseline's 7,500 tok – a 4.8$\times$ in-run reduction (79% fewer tokens). Against the full LongMemEval corpus ($\sim$ 115k tok average), the true reduction is $\sim$ 73$\times$. The 30k-character truncation is not a strawman: it is the operating budget of a commodity edge-window deployment—the 32k context boundary of standard open-weight models—and is applied identically to both arms of the comparison. Full-corpus evaluation across all six question categories is actively queued for the expanded full-length manuscript.

Table 5: E1 decay ablation (E7's n=55, 78 wks, balanced profile): identical streams, only the decay clock differs.

Stream	Metric	τ-decay	Wall-decay
uniform (control)	retention	35/55	35/55
uniform (control)	retrieval	1.00	1.00
burst-quiet (stress)	retention	**35/55**	**0/55**
burst-quiet (stress)	retrieval	**1.00**	**0.00**

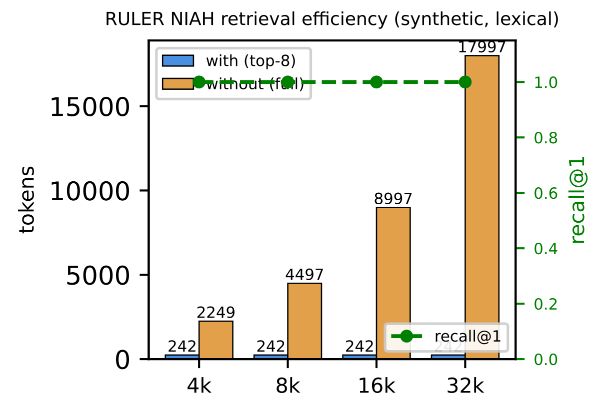

Figure 3: RULER NIAH efficiency (4–32k, synthetic, lexical): flat 242 tok top-8 injection vs. growing full haystack; recall@1 1.0.

Decay ablation (E1): τ-keyed vs wall-clock-keyed decay. The strongest confound test is not retrieval quality but the decay clock itself. We replay identical 55-event streams through two engines that differ **only** in the clock driving the prune ladder (Eq. decay): polytemporal τ (novelty-integrated) vs. wall-clock t (MemoryBank-style $R = e^{-t/S}$). On a uniform stream (control) the two clocks are calibrated to retain identically ($S_{\text{wall}} = 340$ d at n=55); on a burst-quiet stream (all 55 events in the first 2 weeks, then 76 quiet weeks) they diverge completely (Table 5).

After an eventful burst followed by equal wall duration of quiet, wall-clock decay purges the entire burst (0/55) while τ-decay preserves it (35/55, retrievable 1.0): on equal wall time, *wall forgets, τ remembers*. *Takeaway: on burst-quiet streams, wall-clock decay forgets the entire burst* (0/55) *while τ-decay preserves it* (35/55)*—the only difference is the decay clock.* This isolates the decay clock as the causal variable — not the embedding, not the retrieval, not the answer model.

Summary

E7 isolates policy as the causal variable (55/55/35/11); n=50 tests ecological validity; RULER quantifies efficiency. Together they show re-derived τ and salience gating outperform mis-parameterized wall-clock retention on identical input.

Federation and Continuity

Federated brains partition episodes into *private* (local-only embeddings and raw text), *accessed-live* (federated retrieval without import), and *imported* (copied with provenance and decay S). Federation is polytemporal: the time_vector travels with the episode, and recipient-side τ and regime indexes determine ranking, not source wall-clock.

Continuity couples a discontinuous inference core to a continuous sensor substrate (fleet) per D7 (James 1890; Bergson 1910). The sensor stream supplies $d\tau/dt$ during gaps; inference marks witnessed versus reconstructed change (H11–H13). Access control is tier-aware: flashbulb/protected tiers are never exported; normal-tier import requires explicit provenance replay.

Discussion

We make no qualia claim: τ is an operationalization—$d\tau/dt = f(\text{novelty-rate}, \text{prediction-error}, \text{engagement})$, measurable as compression-progress (Schmidhuber 1991) and retrievable via `episode_tau_idx`.

Limitations. τ is proxy-sensitive (the novelty estimator is a choice); escalation bursts require calibrated arousal thresholds (H16); reconstruction grading (H5/H6) can hallucinate schema-consistent details, mitigated by confidence calibration and fuzz-interval reporting.

Validity threats. LongMemEval n=50 covers the *single-session-user* slice of a 500-question, six-type benchmark, and the $acc_{without}$ baseline uses a 30k-character truncated haystack, so the Δ is relative to a weakened baseline rather than full-context recall. RULER $1/10$–$1/74$ at 4–32k may not generalize to 128k. E7 is synthetic policy isolation, not human longitudinal data. The n=50 single-session slice is the proof-of-concept bound; the full 500-question, six-type LongMemEval suite is the targeted next phase, alongside a salience-gated burst variant of the E1 ablation and a continuously-observing fleet deployment for the D7 substrate claim.

Future work pits τ-keyed against wall-clock-keyed memory on decision quality over quiet versus eventful stretches of equal wall duration (D6/H-τ), and tests whether re-derived scheduling (boredom/curiosity) improves consolidation cadence versus fixed windows.

Conclusion

Agents have no felt time, yet we keep giving them memory schedules borrowed from bodies that do. The fix is not to copy human memory wholesale—it is to reuse the mechanisms whose logic transfers and re-derive the rest from what agents actually are. We reuse reconstructive recall, spreading activation, and complementary learning systems; we re-derive time itself, replacing wall-clock decay with subjective τ, salience-set tiers, anchor constellations, and provenance-aware federation.

The re-derivation is measurable, not metaphorical. Retention policy alone, on identical input, yields divergent forgetting (E7: 55/55/35/11). On real LongMemEval the τ-keyed index answers 0.90 vs. 0.12 without it; on RULER it reads a 32k haystack in $1/74$th the tokens. Curation is the hard problem of memory—and it is a design choice, not a model property.

Borrow wheels, not clocks.

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Checklist

1. Claims in abstract match experiments (E7 55/55/35/11, n=50, RULER 1/10–1/74) — Yes. Justification: E7 table/figure and LongMemEval table included; metrics-50.json provided.
2. Limitations described — Yes. See Discussion.
3. No PII; private state in positronic-private excluded — Yes. ENGRAM_TAG=v0.2.0; cs.AI primary.
4. Reproducibility: code, data, retention policies logged — Yes. `figs/e7_survival.pdf` (Task 2), `figs/longmemeval_table.tex` (Task 3), `figs/metrics-50.json`.
5. Citations complete; hyperref forbidden observed — Yes. `aaai2026.sty` unmodified; no hyperref.

Harness Validation

Synthetic n=50 recall@1 1.0 p95 0.7ms validation only (not a claim): in-memory harness with typed polytemporal indexes (episode_tau_idx, episode_fuzz_idx GiST, episode_regime_idx) achieves recall@1 1.0 and p95 0.7ms on 50 synthetic episodes; validation only, excluded from main E7/n=50/RULER claims. Real n=50 LongMemEval metrics in `figs/metrics-50.json` show p95 210s (archival/local, hybrid judge) vs harness 0.7ms, confirming synthetic vs. real gap. Harness verifies Activate·Reconstruct·Decay·Fuse executes over `time_vector` without JSONB scan, and gate-at-encoding H14/H16 logic admits/rejects per salience thresholds.

Hardware and runtime (reproducibility). Host `ai-test` VM (KVM/QEMU, `systemd-detect-virt kvm`, DMI `KVM/Red Hat`): 2× Intel Xeon E5-2698 v4 @ 2.20 GHz (70 vCPU, 2 sockets ×35 cores, 1 thread/core), 64 GiB RAM + 41 GiB swap, 2× AMD Radeon gfx906 (Vega 20) 32 GiB VRAM each (`/dev/dri/card0,1`, `renderD128,129`), ROCm HIP runtime 1.1, `llama-server` 11221 (055fa70af, Clang 17.0.6, `beellama-check/build-hip`). Embedding: `bge-m3-Q8_0.gguf` 606 MiB (`/usr/local/devel/models/embedding/`) served at `:8090` with `--embedding --pooling cls -c 8192` (per-input 512 tok ≈1200 c, `BATCH 8`). LLM: `Qwen3.8-27B-HauhauCS-IQ4_XS.gguf` 15 GiB + draft `Qwen3.8-FastMTP-32K.gguf` 862 MiB + `mmproj HauhauCS-BF16` 889 MiB served at `:8080` with `-t 32 -c 131072 -ngl all --jinja -fa on -ctk q8_0 -ctv q8_0 -ub 2048 --spec-type draft-mtp --spec-draft-ngl all -n-max 4 -p-min 0 --cache-reuse 512`. Kernel `vmlinuz-6.12.96+deb13-amd64`. OpenRouter `meta/muse-spark-1.2-contributor` answers + `meta-llama/llama-3.3-70b-instruct` hybrid judge (`harness/judge.py:39` timeout 60 s, `real_driver.py:89` 120 s). ENGRAM_TAG=`v0.2.0` pinned.

Hypothesis Index (H1–H18)

Compact reference index; each hypothesis is stated where it is used in the main text and sourced from its disanalogy (D*) and case (C*).

- **H1** — Texture-indexed retrieval beats date-range retrieval on human-style probes (D1).
- **H2** — Salience at write-time sets the durability tier; flashbulb-tier is exempt from decay (D2).

- **H3** — “When” is answered by anchor-traversal, not absolute dates (D3, D9).
- **H4** — Repeated near-identical episodes fuse into a schema-trace; only prediction-errors retain episode status; storage grows $\sim \log$ (D4).
- **H5** — Recall is generative reconstruction, graded by confidence; errors cluster as schema-consistent intrusions (D5).
- **H6** — Confidence tracks accuracy (D5).
- **H7** — Memory is hierarchically organized: lifetime periods $\supset$ general events $\supset$ specific knowledge (self-memory system).
- **H8** — Curiosity/boredom (compression-progress) gates encoding cadence (D8).
- **H9** — Positions are fuzzy intervals whose reported width equals encoding confidence (D9, D10).
- **H10** — Activation spreads over associative edges (spreading activation).
- **H11–H13** — A discontinuous inference core coupled to a continuous sensor substrate; change is marked witnessed-vs-reconstructed (D7).
- **H14** — Gate-at-encoding: $d\tau/dt$ advances on surprise only; low-salience episodes never enter the store (D4).
- **H15** — Resolution ladder: schema and episode details fuse at graded resolution (D4, D5).
- **H16** — Escalation bursts temporarily hold the gate open on high-arousal triggers (D2).
- **H17–H18** — Objects condense by prediction-error gating: schema-trace fusion + deferred significance promoting tiers (D4).
- **H-τ** — τ-keyed decay outperforms wall-clock-keyed decay on quiet-vs-eventful equal-duration probes (D6; E1).

RULER Efficiency

RULER efficiency across 4–32k: top-8 retrieval uses a flat 242 tok vs. a linearly growing full haystack (2,249–17,997 tok), a $1/10$–$1/74$ token reduction with recall@1 1.0 at all lengths; full sweep in `figs/ruler_efficiency.pdf`. Object layer: objects condense per H17–H18 via prediction-error gating; escalation bursts H16 hold the gate open on high-arousal triggers; schema-trace fusion merges repeated near-identical episodes (H4/H15 resolution ladder) while retaining prediction-error episodes for detail reinstatement; deferred significance promotes tiers from downstream reference counts. Gate-at-encoding H14 selects durability tier and initial strength S (flashbulb $\Rightarrow$ `Infinity`); `day_class`/`precision_src` lookup types and `provenance` {witnessed,boundary,reconstructed} travel with the polytemporal vector. Recall is Activate·Reconstruct·Decay·Fuse; federation partitions private/accessed-live/imported; regime indexes (`cycle_id`,`phase_idx`) and fuzz `tstzrange` GiST enforce no calendrical primitive.

Engineering recipe. To retrofit polytemporal indexing onto an existing RAG/agent-memory stack: (1) add `tau` (double, NOT NULL), `fuzz` (`tstzrange`), and `mono` (bigint) columns to the episode table, leaving `wall` as a payload attribute; (2) compute `tau` at write-time from the surprise estimator and index it (`episode_tau_idx`); (3) replace wall-clock-ranked decay with strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$; (4) add a GiST index on `fuzz` for interval containment and a (`cycle_id`,`phase_idx`) regime index. No JSONB on any query path.